3. Culture condition

pH

The pH of medium is normally adjusted between 5.6 to 6.0 for the best result.

Temperature

The cultures should be incubated normally at constant temperature of 25°C ± 2°C for optimal growth.

Humidity and Light Intensity

The cultures require 50-60% relative humidity and 16 hours of photoperiod by the illumination of cool white fluorescent tubes of approximately 1000 lux.

Aeration

Aeration to the culture can be provided by shaking the flasks or tubes of liquid culture on automatic shaker or aeration of the medium by passing with filter-sterilized air.

4. Induction of Callus

Explant of 1-2 cm sterile segment selected from leaf, stem, tuber or root is inoculated (transferring the explants to sterile glass tube containing nutrient medium) in the MS nutrient medium supplemented with auxins and incubated at 25°C ± 2°C in an alternate light and dark period of 12 hours to induce cell division and soon the upper surface of explant develops into callus. Callus is a mass of unorganized growth of plant cells or tissues in *in vitro* culture medium.



Figure 5.4: Induction of callus

5. Embryogenesis

The callus cells undergoes differentiation and produces somatic embryos, known as **Embryoids**. The embryoids are sub-cultured to produce plantlets.



Figure 5.5: Embryogenesis

6. Hardening

The plantlets developed *in vitro* require a hardening period and so are transferred to greenhouse or hardening chamber and then to normal environmental conditions.

Hardening is the gradual exposure of *in vitro* developed plantlets in humid chambers in diffused light for acclimatization so as to enable them to grow under normal field conditions.

5.2.3 Types of Plant tissue cultures

Based on the type of explant other plant tissue culture types are

1. Organ culture
2. Meristem culture
3. Protoplast culture
4. Cell suspension culture.

1. Organ culture

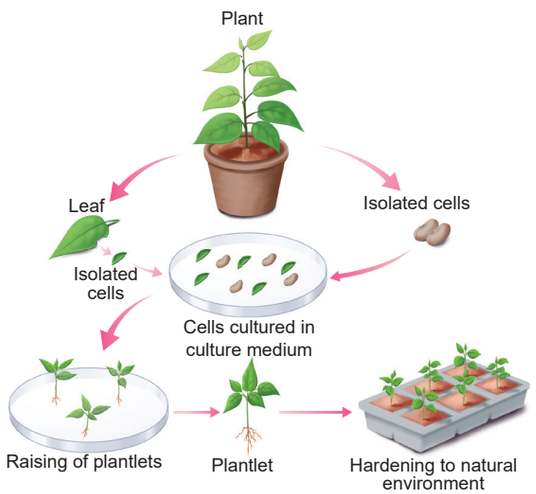


Figure 5.6: Organ Culture

The culture of embryos, anthers, ovaries, roots, shoots or other organs of plants on culture media.

2. Meristem Culture:

The culture of any plant meristematic tissue on culture media.

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